Speedtest Data Pre-processing & Upload Guidelines (SOP)

> **Strict Pre-Processing Rule:** All files **must** be sanitized and realigned based on the **latest official Excel template**. The system will only accept post-processed files. Any unformatted or misaligned files will fail during batch processing.

1. Summary Checklist for Uploaders

Before uploading any file, verify that:

- All columns align exactly with the latest official Excel template.
- Required fields (**colored headers**) are 100% populated with valid data.
- Timestamps reflect **Philippine Standard Time (PHT, UTC+8)** and are strictly formatted as `MM/DD/YYYY, time` (Month first).
- Numbers and decimal fields contain no spaces, commas, or text symbols.
- Dropdown and boolean fields (`signal_quality` , `operator` , `network_type` , `nbc` , `success`) use exact valid values.
- Longitude header is strictly named `lon` (do not rename to `lng` or `long`).
- Rows with unsupported network types (e.g., `Wi-Fi` , `Ethernet`) are removed.
- Empty optional fields are left completely **BLANK** (no `"---"` , `"N/A"` , or `"null"` placeholders).
- Missing readings/unserved locations are marked as `nbc = TRUE` or removed from the file.

2. Required Fields (Colored Headers)

The following 6 columns are **Mandatory Required Fields** (indicated by colored headers in the template). If any of these cells are blank or invalid, file processing will **FAIL**:

Header Name	Type	Description	Action Required if Missing
`timestamp`	DateTime	Test date and time in Philippine Standard Time (`MM/DD/YYYY, time`)	Remove row or fix timestamp format
`upload`	Decimal	Upload speed in Mbps	Remove row or verify reading
`download`	Decimal	Download speed in Mbps	Remove row or verify reading
`operator`	Dropdown	Telecom service provider	Select valid operator from dropdown
`lat`	Decimal	Latitude coordinate (WGS84)	Remove row if missing
`lon`	Decimal	Longitude coordinate (WGS84) — *Note: named* `Lon`	Remove row if missing

> **Caution:** If a row is missing any of these required fields and valid data cannot be retrieved, **delete the entire row** from the spreadsheet before uploading.

3. Timestamp Standardization (MM/DD/YYYY, time)

Timezone Requirement: Philippine Standard Time (PHT, UTC+8)

- Test times must reflect **local Philippine Time** as captured by test handsets or drive-test tools.
- **Do NOT manually convert to UTC:** The QoS processing engine automatically normalizes PHT to UTC on ingestion. Manual subtraction will cause timestamps to be shifted backwards by another 8 hours.

Mandatory Date Format: Month First (MM/DD/YYYY)

The system accepts both 24-hour and 12-hour (AM/PM) time notations, provided the **month comes first**:

- **24-Hour Format:** `MM/DD/YYYY, HH:mm:ss` (e.g., `07/10/2025, 14:30:00` for July 10, 2025, at 2:30 PM)
- **12-Hour Format:** `MM/DD/YYYY, hh:mm:ss tt` (e.g., `10/06/2026, 04:12:00 PM` or `10/06/2026 04:12 PM`)

> ⚠️ **Critical Warning on Date Ambiguity (Month First):** Excel and system date parsers evaluate dates using the `MM/DD/YYYY` standard:
>
> 1. **Invalid Day in Month Position:** A date like `25/06/2025, 08:30` will trigger an immediate **parsing error** because month 25 does not exist.
> 2. **Silent Data Misinterpretation (High Risk):** If a test occurred on **July 10, 2025** and is entered as `10/07/2025`, the system will interpret and store it as **October 07, 2025**.

How to Fix Timestamps in Excel:

1. Highlight the `timestamp` column.
2. Ensure cell values follow `MM/DD/YYYY, HH:mm:ss` or `MM/DD/YYYY, hh:mm:ss AM/PM` format.
3. Review dates where Day ≤ 12 to guarantee that the Month comes first.

4. Numeric & Decimal Value Sanitization

Decimal and numeric fields (e.g., `upload`, `download`, `lat`, `lon`, `rssi`, `ping`, `jitter`) must contain **clean numeric data only**.

- **No Extra Spaces:** Remove leading, trailing, or embedded spaces (e.g., `" 12.34 "` → `12.34`).
- **Use Period (.) for Decimals:** Replace European decimal commas with periods (e.g., `1,25` → `1.25`).
- **No Units/Text Symbols:** Remove units such as `Mbps`, `ms`, `dBm`, or degree symbols (e.g., `15.5 Mbps` → `15.5`).
- **Column Header Distinction:** The longitude column must remain named `lon` (do not rename to `lng` or `long`).

5. Dropdown & Boolean Columns Validation

Dropdown fields must strictly match the pre-approved option list. Custom text or unmapped values will cause file processing to **FAIL**.

Column	Allowed Values	Mapping / Action Required
`signal_quality`	`VeryWeak`, `Weak`, `Fair`, `Good`, `Excellent`	Re-select from the template dropdown list. Do not enter custom descriptions.
`operator`	Registered operators (e.g., `SMART`, `GLOBE`, `DITO`)	Select strictly from the dropdown list. Ensure capitalization matches.

Column	Allowed Values	Mapping / Action Required
`network_type`	`NR` , `LTE` , `TDDLTE` , `FDDLTE` , `HSPAP`	Map raw values: • 5G → `NR` • 4G → `LTE` , `TDDLTE` , `FDDLTE` • 3G → `HSPAP` • **Wi-Fi / Ethernet → ❌ Remove entire row**
`nbc`	`TRUE` , `FALSE` , `NBC` (or leave BLANK)	• Set to `TRUE` or `NBC` if no broadband coverage is detected. • Set to `FALSE` or leave `BLANK` for normal active tests.
`success`	`TRUE` , `FALSE` (or leave BLANK)	• Set to `TRUE` if test execution completed successfully. • Set to `FALSE` if test failed or aborted.

> **Remove Non-Mobile Rows:** If `network_type` is `Wi-Fi` , `Ethernet` , or any non-cellular network, **delete the entire row** from the file.

6. Blank Values vs. Dummy Placeholders

For non-required columns without data:

- Leave the cell completely **BLANK (EMPTY)**.
- **DO NOT** type placeholder characters such as `"---"` , `"-"` , `"N/A"` , `"null"` , `"NONE"` , or `"0"` (unless zero is an actual measured value).

7. Handling Zero Readings vs. NULL (NBC Logic)

Scenario	Definition	System Action
Valid Zero Reading	Test was successfully executed, but measured speed was `0.00 Mbps` upload and download.	**Include in file:** Set `upload = 0` and `download = 0` .
No Reading / NULL / Unserved Area (NBC)	Test was not performed, signal was unavailable, or reading returned `NULL` .	**Mark `nbc = TRUE` (Recommended):** The system automatically records zero throughput and registers the point for No Broadband Coverage gap analysis. *(Alternatively, remove row from file).*

8. Post-Upload Support

If your file encounters processing errors even after following this guide:

1. Record the exact row and column number specified in the error message dialog.
2. Verify the template alignment and field types for that row.
3. Confirm that timestamps follow the `MM/DD/YYYY` month-first standard in Philippine Time.
4. Contact the system administrator with the error log for further assistance.